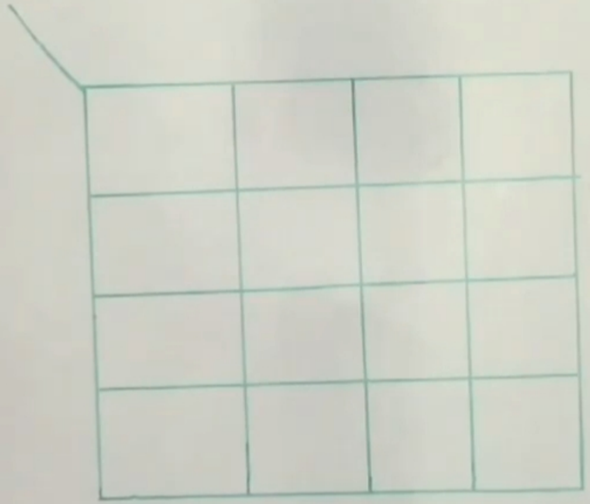
4-Bit Even Parity Generator

B_3	B_2	B_1	B_0	P
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0



=> **Generates parity 1 for odd number of 1s in input**

4-Bit Even Parity Generator

B_3	B_2	B_1	B_0	P	
0	0	0		0	(0)
0	0				(1)
0	0				(2)
0					(3)
0					(4)
0				0	(5)
0				0	(6)
				1	(7)
					(8)
					(9)

B_1B_0 \ B_3B_2	00	01	11	10
00	0	1 4	12	1 8
01	1 1	5	1 13	9
11	3	1 7	15	1 11
10	1 2	6	1 14	10

Check-Board Configuration

$$\boxed{P = B_0 \oplus B_1 \oplus B_2 \oplus B_3}$$

$$P = \bar{B}_3\bar{B}_2\bar{B}_1B_0 + \bar{B}_3\bar{B}_2B_1\bar{B}_0 + \bar{B}_3B_2\bar{B}_1\bar{B}_0 + \bar{B}_3B_2B_1B_0 + B_3B_2\bar{B}_1B_0 + B_3B_2B_1\bar{B}_0 + B_3\bar{B}_2\bar{B}_1\bar{B}_0 + B_3\bar{B}_2B_1B_0$$

$$P = \bar{B}_3\bar{B}_2(\bar{B}_1B_0 + B_1\bar{B}_0) + \bar{B}_3B_2(\bar{B}_1\bar{B}_0 + B_1B_0 + B_3B_2(\bar{B}_1B_0 + B_1\bar{B}_0) + B_3\bar{B}_2(\bar{B}_1\bar{B}_1 + B_1$$

$$= \left[\bar{B}_3\bar{B}_2(B_0 \oplus B_1) + \bar{B}_3B_2(\overline{B_0 \oplus B_1}) + B_3B_2(B_0 \oplus B_1) + B_3\bar{B}_2(\overline{B_0 \oplus B_1})\right]$$

$$= (B_0 \oplus B_1)(\bar{B}_3\bar{B}_2 + B_3B_2) + \overline{(B_0 \oplus B_1)}(\bar{B}_3B_2 + B_3\bar{B}_2)$$

$$= \underset{A}{(B_0 \oplus B_1)}\,\underset{\bar{B}}{(\overline{B_2 \oplus B_3})} + \overline{(B_0 \oplus B_1)}\,(B_2 \oplus B_3)$$

$$+ \bar{A} \cdot B$$

$$= A \oplus B.$$

$$\boxed{P = B_0 \oplus B_1 \oplus B_2 \oplus B_3}$$